

EVALUATION OF LAND RESOURCES IN PAGERU RIVER BASIN BY USING REMOTE SENSING DATA

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ABSTRACT

The land use/land cover classification of Pageru river basin was done with the help of IRS-1B-LISS II FCC on 1: 50,000 scale by visual interpretation techniques and in conjunction with SOI topo sheets (57 J/6, 7, 10 and 11). Based on the remote sensing data, the land use/land cover patterns were interpreted. The built-up land in the basin area is only 1.02 per cent, while agricultural land occupies a large area covering 72.92 per cent. The double cropped area is very limited occupying 3.18 per cent only. The area under degraded/open forest occupy only 3.99 per cent and the waste lands occupy 14.42 per cent. The area occupied by surface water bodies is only 1.97 per cent and the grass land/grazing land occupies 2.04 per cent. Even though the agricultural land available in the basin area is substantial, the total land could not be brought into cultivation due to meagre surface and groundwater resources present in the catchment area.

INTRODUCTION

Land use refers to "man's activities and various uses which are carried on land". Land cover refers to "natural vegetation, water bodies, rock/soil, artificial cover and others resulting due to land transformation" (Stamp, 1962). Although land use is generally inferred based on the cover, yet both the terms land use and land cover are closely related and interchangeable. The interactions between different land uses on different parcels of land are ever changing and it is essentially dynamic (Gautam and Narayan, 1982).

In the present study, the land use/land cover map has been prepared using remote sensing data in order to generate basic information on land use.

MATERIAL AND METHODS

Study Area: Pageru, a tributary of Papaghni river rises from Seshachalam hill ranges and flows easterly to join at the confluence of Pennar and Papaghni rivers at Putlampalli village near Kamalapuram in Cuddapah district. The catchment area of the basin is 480 km² and is covered by Survey of India toposheet numbers 57J/6,7,10 and 11 (1:50,000). The basin area lies in between

north latitudes 14° 29' 04" and 14° 39' 51" and east longitudes 78° 19' 12" and 78° 41' 32" (Fig. 1).

Geology: The rock formation of the study area are classified as Cuddapah super group and Kurnool group. The oldest are Tadipatri formations belonging to lower Cuddapah super group composed of slaty shales with interbedded thin layers of siliceous limestone, quartzite and basic intrusive sills. The Tadipatri formations are overlain by the rocks of the Kurnool group which include quartzite, limestone and shale. Alluvium of holocene and recent age is more or less stratified deposit of gravel, sand, silt, clay occurring all along the course of the stream. The alluvium at the confluence near Putlampalli is found to comprise sand of variable thickness and is composed of sandy aquifers occupying the easternmost part of the study area (Fig. 2).

Climate: The general climate of Pageru river basin is semi-arid in nature. The temperature reaches high in the month of April and it is low in the month of December. The temperature gradually rises from the month of January and reaches maximum in the month of April and decreases gradually from May and

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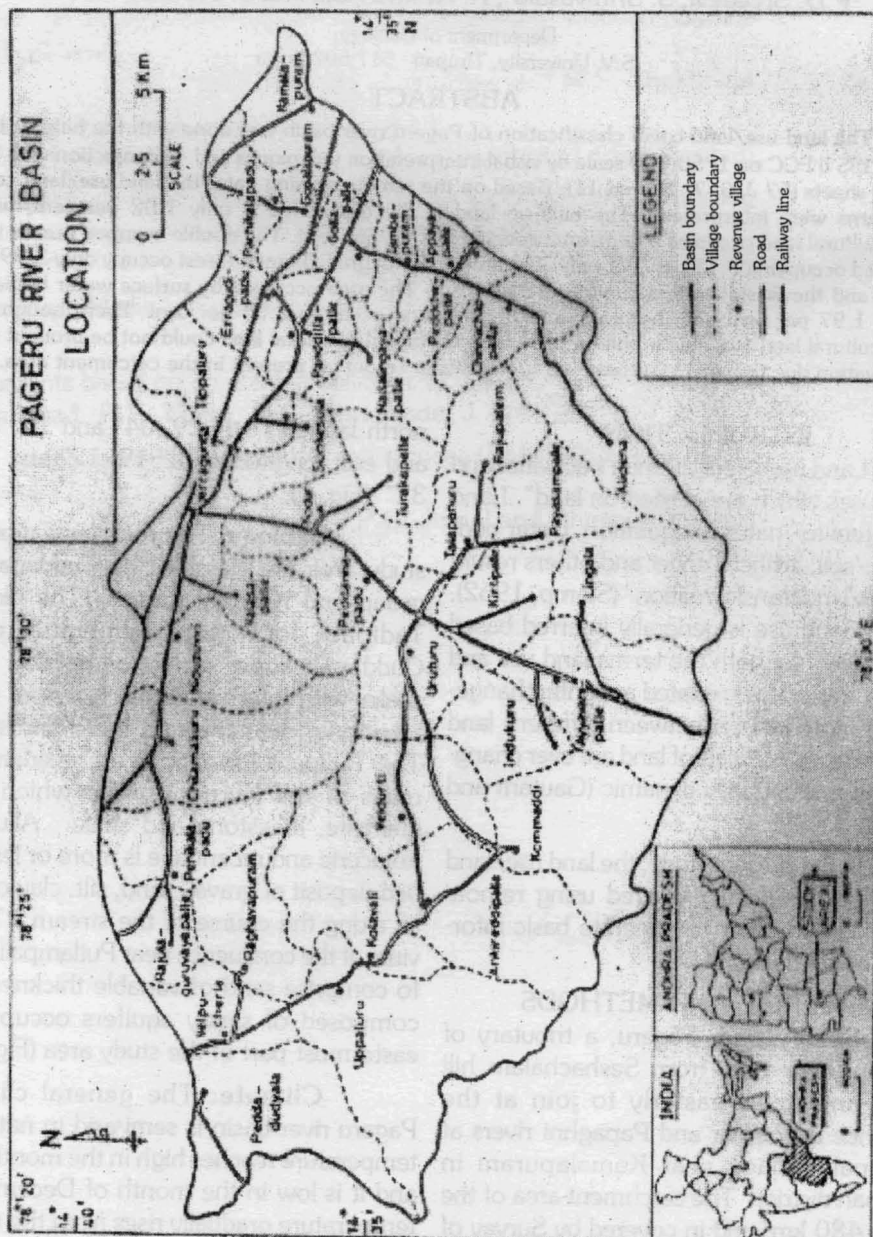


Fig. 1. Location

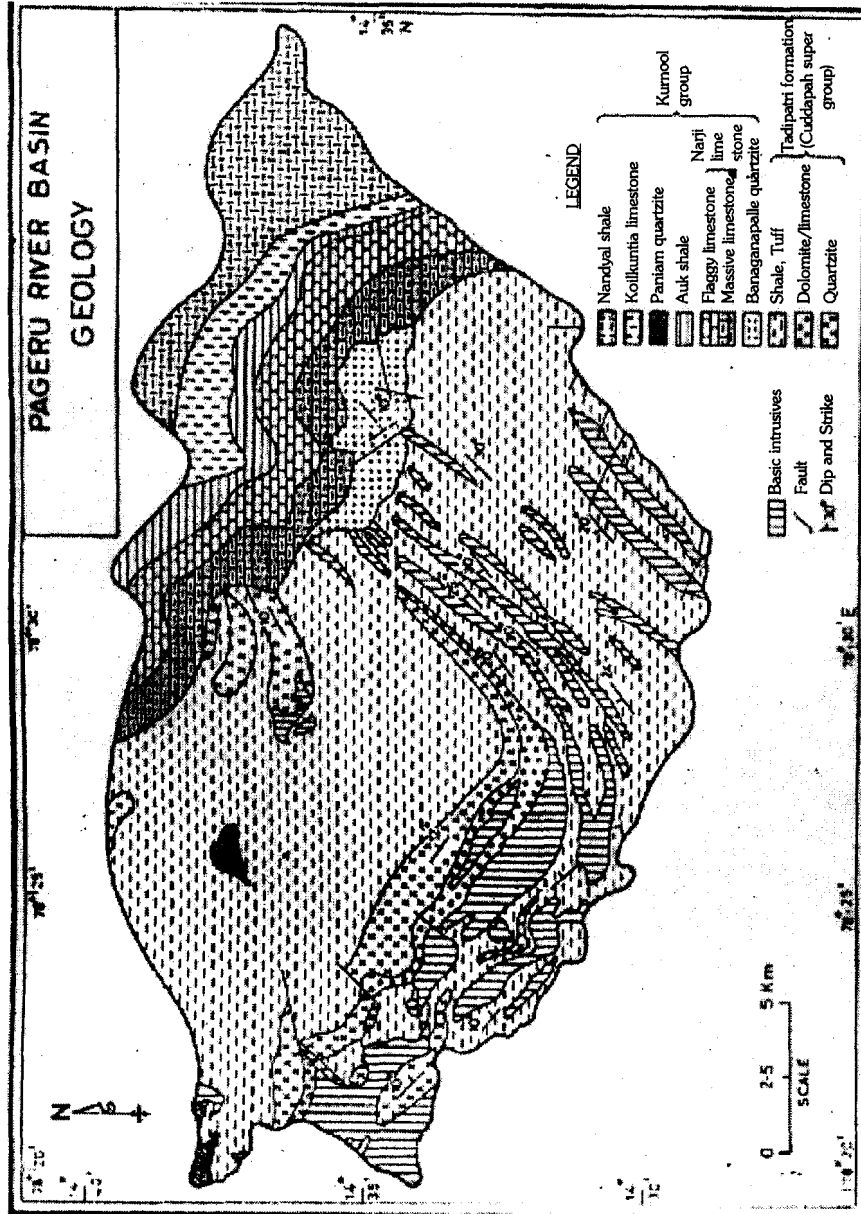


Fig. 2. Geology

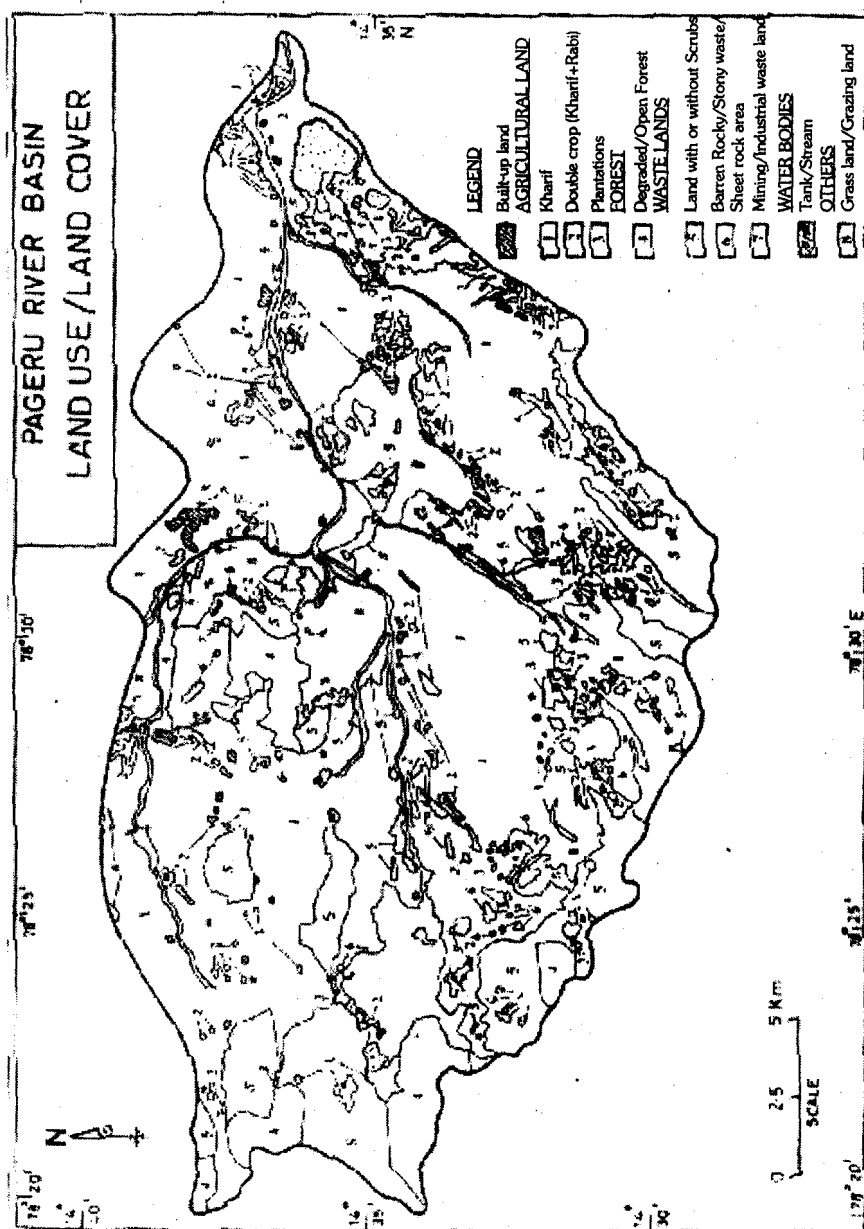


Fig. 3. Land use/Land cover

reaches minimum in the month of December. The relative humidity shows seasonal variations generally fluctuating with rainfall and temperature apart from the diurnal variations. The relative humidity is more during monsoon and winter period due to increase in rainfall and decrease in temperature. The relative humidity is less in summer due to the combined effect of high temperature and low rainfall. The Pageru river basin receives an average annual rainfall of 580.79 mm. with significant seasonal variations. Usually the region receives its first spell of rainfall from pre monsoonal convectional showers in the month of May. But its occurrence is erratic. The intensity and amount of rainfall is unpredictable during the southwest monsoon period (June to September). Infact, highest rainfall occurs in the basin during the Northeast monsoon period (October

and November). The period between January and May is the main dry season and receives some rainfall due to convections or winter cyclonic disturbances.

The land use/land cover classification is done with the help of IRS-1B, LISS - II FCC Satellite images for both *kharif* and *rabi* seasons Geo coded at 1:50,000 scale by visual interpretation techniques and in conjunction with SOI topo sheets Nos. 57 J/6,7,10 and 11. The land use/land cover classification system developed by the National Remote Sensing Agency, Hyderabad, later modified on the suggestions made by the Planning Commission of India has been adopted in the present study (NRSA, 1990). The system of classification and criteria used for Identifying various land use/land cover classes is given in Table 1.

Table 1. Land use/Land cover classification system

Level - I	Level - II	Level - III	Symbols
1. Built - upland	1.1 Villages		
2. Agricultural land	2.1 Cropland +	2.1.1 Kharif	01
		2.1.2 Kharif + Rabi (Doubled cropped)	02
	2.2 Plantations #		03
3. Forest	3.1 Degraded	3.1.1 open	04
4. Waste lands	4.1 Land with or with out scrub		05
	4.2 Berren rocky/stony waste/ sheet rock area		06
	4.3 Mining/Industrial Waste land		07
5. Water bodies	5.1 River/Stream		
	5.2 Tank		
6. Others	6.1 Grass land/grazing land		08

+ It includes land under agricultural crops during *Kharif*, (both irrigated + unirrigated) and the area under double crop (cultivated during *Kharif* as well as *Rabi* seasons)

It includes all agricultural plantations like coconut, citrus and other horticultural crops.

RESULTS AND DISCUSSION

The detailed land use/land cover statistics of the Pageru river basin for the year

1998 is given in Table 2 and their spatial distribution (Fig. 3).

Table 2. Land use/Land cover classification

	Area in km ²	Percentage
Built-up land	4.88	1.02
Agricultural land		
Kharif	349.98	72.92
Double crop	15.25	3.18
Plantations	2.25	0.46
Forest		
Degraded/open forest	19.13	3.99
Waste lands		
Land with or without scrub	63.50	13.23
Barren rocky/stony waste/shoet rock area	4.13	0.86
Mining/industrial waste land	1.56	0.33
Water bodies		
River/stream	4.78	0.99
Tank	4.73	0.98
Others		
Grass land/grazing land	9.81	2.04

BUILT-UP LAND

It is defined as an area occupied by human habitation developed due to non-agricultural use and which has a cover of buildings, transport, communications and utilities in association with water, vegetation and vacant lands.

The physical size or built-up sprawl with transport network is used to classify settlements as urban or rural. It is usually identified on the imagery by its dark bluish green to bluish tone, definite size, shape, and texture. Often, built-up land with high density of buildings etc., appears in dark tone at the centre and lighter on the peripheries, due to low density. The pattern is contiguous to non-contiguous (punctuated by vacant lands and vegetation), clustered or scattered. Usually the built-up area lies in all types of terrain amid agricultural lands, forests and waste lands in association with road, rail and canal networks and artifacts. They appear distinctly in between the crop lands of *kharif* and *rabi* seasons. As there are no urban areas, the rural settlements present in the study area cannot be identified in satellite imageries as they are very small in extent comprising a total area of about 4.88 km², accounting 1.02 per cent to the study

area (Calculated with the help of SOI topo sheets Nos. 57 J/6,7,10 and 11).

AGRICULTURAL LAND

It is defined as the land primarily used for farming and for production of food, fibre, other commercial and horticultural crops. It also includes land under crops, plantations etc.

The tonal contrast of crop land varies from bright red to red that may signify greenness of the foliage at different stages of crop growth and phenological condition (healthy or infected) besides the nature of soil (moist or dry), the type of terrain etc. Its spatial extent varies in size and shape, with smooth texture (when the crop is in full matured stage) to coarse or mottled (at the early stages of planting and growth). It is contiguous under irrigated (canal, tank or well etc.) areas and noncontiguous in unirrigated or rain-fed dry lands. Very often contiguity of croplands is punctuated by harvested fields or fallow lands. Crop lands occur on a variety of terrain, often in association with the terrain patterns as observed in river plains, valleys, coastal lowlands etc. Cropping season and cropping pattern (single, double or mixed) vary from season to season, terrain conditions, farming and tenural systems, occurrence and distribution of rainfall etc. The

major crop cultivated in the study area is groundnut. Crops like paddy, cotton, pulses, sunflower are grown in Kharif and Rabi seasons with the help of supplemental irrigation.

The largest agricultural land use category in the basin is rainfed area occupying 349.98 km², during kharif season accounting for 72.92 per cent. The double cropped area in the basin is 15.25 km², accounting for 3.18 per cent only.

Plantations: It is described as an area under agricultural tree crops planted adopting certain agricultural management techniques. These include coconut, citrus and other horticultural crops. The plantations appear in imageries with dark red to red in tone, small in size with regular shapes, sharp and smooth edges. The difference in tone may signify different types of plantations or same plantation in different stages of growth or foliage cover or due to seasons. These occur on both uplands, and foot hills and occasionally on river plains also. The area under plantations covers of about 2.25 km², accounting 0.46 per cent to the study area.

FOREST

It is an area bearing an association predominantly of trees and vegetation types capable of producing timber and other forest products.

These appear bright red to dark red in tone in all the months of the year because of their foliage, very dense canopy associated with moisture all-round the year. They vary in size and shape, and are smooth in texture and contiguous in spread but occasionally noncontiguous due to forest clearing. These occur on the higher reaches of hill-slopes coinciding with zones of very high rainfall.

Degraded forest: It is described as a forest, where the vegetative density is less than 10-40 per cent of the canopy cover. It is the result of both biotic and abiotic influences. It

appears light red to dark brown in tone subject to the amount to foliage cover and season. The areal spread varies in size with irregular and discontinuous shapes. Degraded/open forests occur on mountain/hill slopes, isolated hills, uplands. The degraded/open forests cover an area of about 19.13 km², accounting 3.99 per cent to the study area.

WASTE LANDS

This is described as degraded land which can be brought under vegetative cover with reasonable effort, and which is currently under utilized land deteriorating due to lack of appropriate water and soil management or on account of natural causes. Waste lands can result from inherent/imposed constraints such as location, environment, chemical and physical properties of the soil and financial management constraints. Waste lands identified in the study area are described below.

Land with or without scrub: They occupy relatively uplands or high ground with or without scrub. These lands are generally prone to degradation or erosion. These exclude hilly and mountainous terrain. They appear in light yellow to brown to greenish blue tone (subject to surface moisture and plant cover). Wherever the occurrence of scrub on the uplands is more, reddish tint appears in dots and patches. They vary in size with irregular and discontinuous shapes and have coarse to mottled texture (subject to surface lithology and vegetation cover on the top) and contiguous and dispersed in patches. Land with or without scrub occupies an area of about 63.50 km², accounting 13.23 per cent to the study area.

Barren/Rocky/Stony sheet rock area: It is defined as the rock exposures of varying lithology often barren and devoid of soil cover and vegetation. They occur amid hill forests as openings or scattered as isolated exposures or loose fragments, boulders or as sheet rocks on plateaus and plains.

This appears in greenish blue to yellow to brownish tone (subject to varying rock type), vary in size with irregular and discontinuous pattern. It occurs in steep hill slopes, crest, isolated hillocks, plateau and eroded plains in association with barren and exposed rocks. Barren rock outcrops cover an area of about 4.13 km², accounting for 0.86 per cent to the study area.

Mining/Industrial waste: Mining and industrial waste cover occupies a small area of about 1.56 km², accounting 0.33 per cent for the basin area. The mining activity is noticed in the northern part of the study area.

WATER BODIES

This class comprises areas of surface water either impounded as ponds, lakes, reservoirs or flowing bodies as streams, rivers, canals etc. These are clearly seen on satellite false colour imagery in blue colour.

River/Stream: It is a natural course of water flowing on the land surface along a definite channel. It may be seasonal or perennial. River/Stream appears in light blue to dark blue tone (subject to shallow surface water spread, deep and more volume of water, turbidity etc.), long and narrow to wide in size with irregular and sinuous shape, smooth to medium in texture (in case of high turbidity), contiguous, nonlinear to dendritic/sub-dendritic pattern. The river/streams occupy an area of about 4.78 km², accounting for 0.99 per cent of the study area.

Tanks: These water bodies appear in light blue to dark blue tone (subject to shallow surface water spread, volume of water, turbidity etc.). The presence of weed/vegetation contributes to patches of red tone amidst them. They are small/medium to large with regular to irregular shapes, smooth to mottled in texture (subject to vegetation cover) noncontiguous and dispersed in pattern, except canals which show a linear pattern. Tanks occur

in lowlands of plains, uplands and valleys. The tanks occupy only 4.73 km², in the basin area accounting for 0.98 per cent to the study area.

Others: It includes all those which can be treated as miscellaneous because of their nature of occurrence, physical appearance and other characteristics. The others identified in the study area are described below.

Grass Land/Grazing Land

It is an area of land covered with natural grass along with other vegetation, often grown for fodder to feed cattle and other animals.

Grass land/grazing land appears light red to light brown in tone, vary in size with irregular shapes, coarse to mottled texture (subject to amount of vegetation) contiguous to non contiguous in pattern. It occurs on uplands, hill slopes, foot slopes, (Piedmont zones) inland plains and close to river/stream beds in association with unirrigated and dry farming agricultural lands, lands fenced from cultivation, pasture grounds and meadows etc. The grass land/grazing land occupies an area of about 9.81 km², accounting 2.04 per cent of the study area.

CONCLUSION

Based on the remote sensing data, the land use/land cover patterns were interpreted. The built-up land in the basin area is only 1.02 per cent, while agricultural land occupies a large area covering 72.92 per cent. The double cropped area is very limited occupying 3.18 per cent only. The area under degraded/open forest occupies only 3.99 per cent and the waste lands occupy 14.42 per cent. The area occupied by surface water bodies is only 1.97 per cent and the grass land/grazing land occupies 2.04 per cent. Even though the agricultural land available in the basin area is more, the total land could not be brought into cultivation due to meagre surface and groundwater resources present in the catchment area.

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